

UNIVERSITY OF CAPE TOWN
DEPARTMENT OF STATISTICAL SCIENCES - STA2020S: APPLIED STATISTICS
CLASS TEST 1 – 12 March 2025

Time: 90
MINUTES

Max Marks: 50

Internal Miss G.
examiners: Carmichael

Instructions:

Answer ALL questions.

Marks are awarded for intermediate calculations.

Answer all MCQ questions *IN PENCIL* on the MCQ sheet provided.

Answer each written question on a *NEW PAGE* in your answer book.

ROUND ALL ANSWERS TO 2 DECIMAL PLACES UNLESS OTHERWISE SPECIFIED.

DO NOT ROUND INTERMEDIARY CALCULATIONS.

SECTION A: MCQ QUESTIONS
[12]

1. Which one of the following goals can we **not** achieve when conducting correlation analysis with two variables? (2)
 - A. Quantifying the strength of a linear relationship between 2 variables.
 - B. Finding the direction of the linear relationship between 2 variables.
 - C. Performing statistical inference to determine whether there is a relationship between 2 variables.
 - D. Predicting one variable's value from the other variable's value.
 - E. Calculating the proportion of variability in one variable that is explained by the other.

2. Which of the following statements about the Akaike Information Criterion (AIC) is correct? (2)
 - A. AIC is used to compare models fitted to different data.
 - B. A higher AIC value indicates a better model fit.
 - C. The AIC of a model measures the same thing as the coefficient of determination (R^2).
 - D. The AIC is a trade-off between model complexity and goodness-of-fit.
 - E. The AIC always increases when additional independent variables are added to the model.

3. Which of the following gives the distribution of the test statistic for a test of significance of the model coefficients in a multiple linear regression model? (2)
 - A. t_{n-p-1}
 - B. $N(0,1)$
 - C. t_{n-2}
 - D. $N(0, \sigma^2)$
 - E. $F_{p,n-p-1}$

4. The following regression model is fitted to data with 30 observations:

$$\hat{y} = \beta_0 + \beta_1 x$$

Using the model, the predicted value of y when $x = 0.9$ is 2.23. The mean of x is 1.1, and the sum of squares for x is 7.71. The residual standard error of the model is 0.96. What is the 95% confidence interval for this prediction? (2)

- A. [0.31, 4.15]
 - B. [0.57, 3.89]
 - C. [1.91, 2.55]
 - D. [0.23, 4.23]
 - E. [1.84, 2.62]
5. We use a multiple linear regression model when: (2)
- A. There is a single continuous dependent variable and a single continuous independent variable.
 - B. There is a single binary dependent variable and multiple continuous/categorical independent variables
 - C. There is a single continuous dependent variable and multiple continuous/categorical independent variables.
 - D. There are multiple continuous/categorical dependent variables and a single continuous independent variable.
 - E. There are multiple continuous dependent variables and a single continuous independent variable.
6. Increasing the probability threshold for a classification from 0.5 to 0.7 would result in: (2)
- A. An increase in the sensitivity of the classification.
 - B. An increase in the specificity of the classification.
 - C. A decrease in the number of true negatives.
 - D. An increase in the number of false positives.
 - E. An increase in the number of true positives.

SECTION B: REGRESSION QUESTIONS

[38]

QUESTION 1

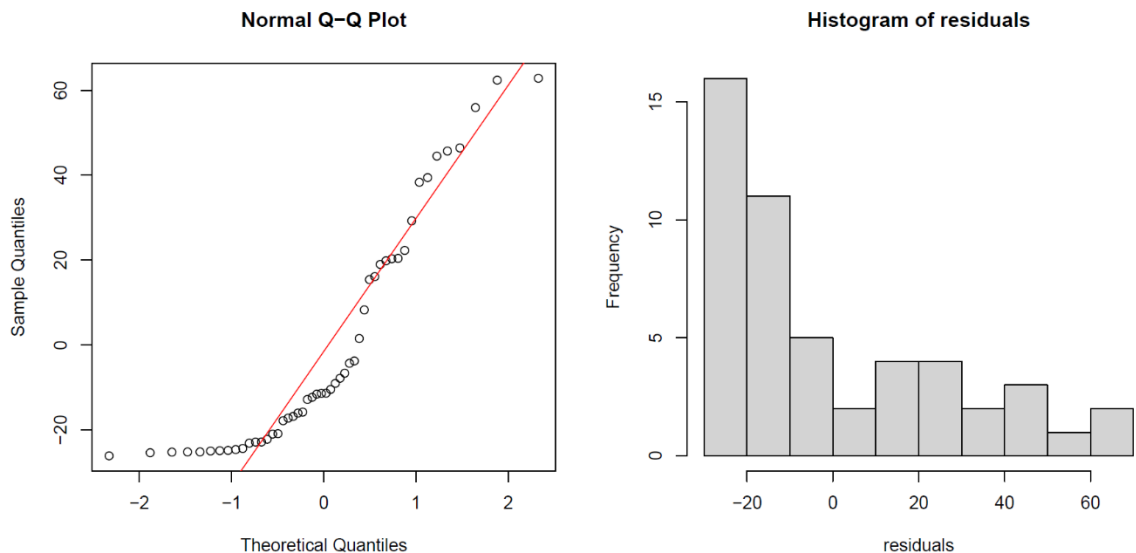
[10]

- a. In a linear regression analysis, why do we conduct a hypothesis test to test the significance of the model coefficients? Refer to the concept of a population versus a sample in your answer. (2)

The model coefficients are calculated using this sample data (1), so we use hypothesis testing to determine whether the relationships we observe in the sample are likely to be present in the population (1). **OR** Award only (1) if they answer along the lines of: to test whether the estimated model coefficients are significantly different from zero.

- b. Examine the model checking plots below. What assumption are these plots checking? Is this assumption met? Justify your answer with evidence from the plots. (4)

These plots are checking the assumption that the model residuals are normally distributed (1). This assumption is not met in these plots (1). In the qq-plot the residuals deviate from the line (1), and the distribution of the residuals in the histogram is very positively skewed (1).



- c. Use the following results from a simple linear regression analysis to calculate the 95% confidence interval for the slope coefficient. The analysis was done for 40 observations.

(2)

$$CI = \hat{\beta}_1 \pm t_{\alpha/2, n-2} \times se(\hat{\beta}_1)$$

$$CI = 4.9234 \pm 2.024 \times 0.1814 \quad (\frac{1}{2} \text{ mark for correct t-value, } \frac{1}{2} \text{ mark for correct } \hat{\beta}_1 \text{ and } se(\hat{\beta}_1))$$

$$CI = [4.56, 5.29] \quad (\frac{1}{2} \text{ mark for correct lower, } \frac{1}{2} \text{ mark for correct upper})$$

Coefficients:				
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.3077	0.3867	13.72	2.64e-16 ***
x	4.9234	0.1814	27.15	< 2e-16 ***

- d. Using a 5% significance level, what conclusion would you make about the significance of the coefficient for x from the model in question 1.c? Justify your conclusion using a p-value. (2)
- p-value $< 2 \times 10^{-16} < 0.05$ ($\frac{1}{2}$) therefore, we reject H_0 ($\frac{1}{2}$) and conclude that there is a linear relationship ($\frac{1}{2}$) between the dependent variable and x ($\frac{1}{2}$).
- Also accept if student says something like p-value $< 0.01 < 0.05$ rather than stating the actual p-value.

QUESTION 2

[12]

An airport wants to avoid flight delays, so to better understand associations with flights being delayed, they collect the following data on 100 flights from the airport:

- *delay_* – flight delay time (in minutes).
- *congestion* – the number of planes arriving or departing the airport within an hour of the flight time.
- *age* – the age of the plane (in years).
- *weather* – weather conditions at take-off time (either “clear” or “not clear”).

A multiple linear regression model is fitted using *delay* as the dependent variable and all other variables as independent variables. The results of this analysis are shown below:

```

Call:
lm(formula = delay ~ congestion + age + weather)

Residuals:
    Min       1Q   Median       3Q      Max
-14.6491  -3.6642   0.2495   3.4820  12.3771

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.06741    2.10946  -0.032  0.97457
congestion     1.15568    0.13096   8.825 4.97e-14 ***
age           0.44322    0.15635   2.835  0.00559 **
weathernot clear 2.96079    1.17048   2.530  0.01305 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.54 on 96 degrees of freedom
Multiple R-squared:  0.4855,    Adjusted R-squared:  0.4694
F-statistic: 30.19 on 3 and 96 DF,  p-value: 7.791e-14

```

- a. Give the regression equation for this model. Provide the actual estimated values for the model coefficients from the given RStudio output. (1)

$$\widehat{\text{delay}} = -0.07 + 1.16 \times \text{congestion} + 0.44 \times \text{age} + 2.96 \times \text{weather not clear}$$

No marks for not using the actual values.

Does not matter if they round off the values or not.

Subtract ½ a mark if they make 1 mistake. If more than 1 mistake, they get 0 for the question.

- b. What is the interpretation of the “weathernot clear” coefficient? Be as specific as possible. (2)

On average ($\frac{1}{2}$), the delay time is 2.96 (2.96079) minutes longer ($\frac{1}{2}$) when the weather conditions are not clear compared to when they are clear ($\frac{1}{2}$), holding all else constant ($\frac{1}{2}$).
Subtract a half mark for not including minutes (don't accept “units”, they need to be specific).

- c. What is the predicted delay time (in minutes) when the congestion is 10 planes, the plane age is 4 years, and the weather condition is clear? (1)

$$\hat{y} = -0.06741 + 1.15568(\text{congestion}) + 0.44322(\text{age}) + 2.96079(\text{weather})$$

$$\hat{y} = -0.06741 + 1.15568(10) + 0.44322(4) + 2.96079(0)$$

$$\hat{y} = 13.26 \text{ minutes (1)}$$

- d. Conduct the F-test to test the significance of the overall model. State both hypotheses, report the p-value of the test, and make a conclusion. You should use a significance level of 5%. (3)

$$H_0: \beta_1 = \beta_2 = \beta_3 = 0 \quad \left(\frac{1}{2}\right)$$

$$H_a: \text{at least one } \beta_j \neq 0 \quad \left(\frac{1}{2}\right)$$

$$\text{p-value} = 7.79 \times 10^{-14} \quad \left(\frac{1}{2}\right)$$

conclusion: $p\text{-value} < 0.05$ ($\frac{1}{2}$) therefore, we reject H_0 ($\frac{1}{2}$) and conclude that the overall model is significant ($\frac{1}{2}$). (Could also say: conclude that there is a linear relationship between delay and at least one of the independent variables)

- e. Before the analysis was conducted, a correlation matrix was produced. Why would the correlation matrix need to be assessed before conducting the analysis? (1)

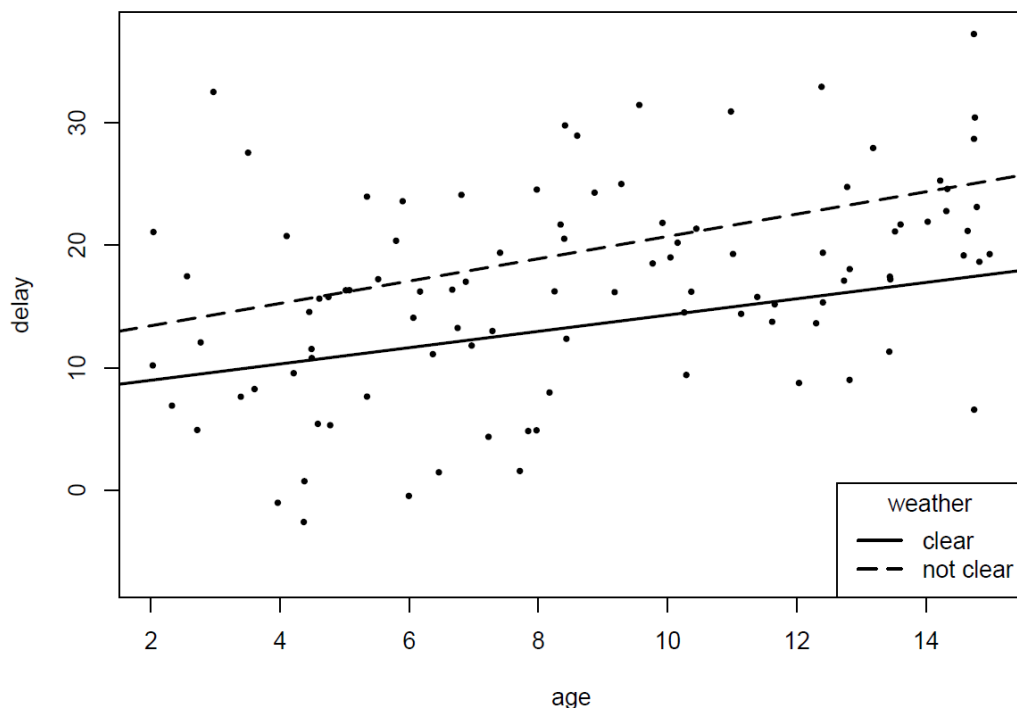
We check correlations between the independent variables to ensure that there is no multicollinearity in our models.

- f. Given that the correlation between *age* and *congestion* was found to be 0.06, would you conclude that the results of the fitted model are problematic? Justify your answer. (1)

This is not an issue ($\frac{1}{2}$) because 0.06 is a very small correlation ($\frac{1}{2}$).

- g. The plot below shows the relationship between *delay* and *age* of the plane. The lines are the fitted regression lines for when the weather is clear (solid line) and for when the weather is not clear (dashed line). By examining this plot, would you conclude that an interaction between *age* and *weather* is likely? Justify your answer. (2)

An interaction between *age* and *weather* is unlikely. (1) The slopes of the two lines are very similar / the two lines look parallel (1).



- h. When do we use a confidence interval versus a prediction interval for a prediction in a regression analysis? (1)

We use a confidence interval when we are predicting the average response and a prediction interval when we are predicting a specific response.

QUESTION 3

[6]

The airport scheduling staff now want to be able to predict flight delays, but they don't know which variables to include in their model. They conduct a model building procedure to help them decide. The results of this procedure are shown below:

```
Start: AIC=346.31
delay ~ congestion + age + weather

              Df Sum of Sq    RSS    AIC
<none>                2946.3 346.31
- weather            1    196.38 3142.7 350.77
- age                1    246.64 3193.0 352.35
- congestion         1   2390.00 5336.3 403.71

Call:
lm(formula = delay ~ congestion + age + weather)

Coefficients:
(Intercept)      congestion            age  weathernot clear
      -0.06741           1.15568           0.44322           2.96079
```

- a) Which model selection procedure has been implemented here? Justify your answer. (1)
Backward elimination ($\frac{1}{2}$). Variables are removed at each iteration (indicated by the minus sign before each variable) ($\frac{1}{2}$).
Can also say that we start with the full model for the last half mark.
- b) Why can it be dangerous to just rely on the results of model building procedures like the one implemented above? (Give 2 reasons) (2)
This can lead to overfitting (capitalising on chance features of the sample) (1). These methods do not incorporate any practical knowledge about the data (1).
- c) Explain the algorithm behind the forward selection model building procedure. (3)
It starts with the intercept model in the first iteration ($\frac{1}{2}$). At each iteration the effect of adding each possible explanatory variable from the model independently is assessed ($\frac{1}{2}$) by calculating the AIC of the resulting models ($\frac{1}{2}$). The variable whose inclusion decreases the AIC the most is added to the starting model in this iteration ($\frac{1}{2}$). This model now becomes the new starting model ($\frac{1}{2}$). This is repeated until no further improvement to the AIC is possible ($\frac{1}{2}$).

QUESTION 4

[10]

A pharmaceutical company wants to determine whether a patient's age will affect whether they respond positively to a new medication the company is developing. They collected data on 100 patients who were prescribed the medication. The dependent variable is binary and takes on a value of 1 if the patient responds positively to the medication and 0 if they do not. Age is recorded in years. The result of fitting a logistic regression model to this data is shown below:

```
Call:
glm(formula = response ~ age, family = "binomial")

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.4457  -0.9023  -0.6026   1.1280   2.0070

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.31075    0.65537   2.000  0.04550 *
age         -0.05880    0.01878  -3.132  0.00174 **
---

```

- a. What is the interpretation of the age coefficient in terms of odds? (2)
 On average ($\frac{1}{2}$), the odds of a patient responding positively change by a factor ($\frac{1}{2}$) of 0.94 ($\frac{1}{2}$) for a 1-year increase in age ($\frac{1}{2}$).
- b. What is the predicted probability of a patient who is 30 years old responding positively to the medication? (2)

$$\text{logodds} = 1.31075 - 0.05880(\text{age})$$

$$\text{logodds} = 1.31075 - 0.05880(30)$$

$$\text{logodds} = -0.45325 \text{ (1 for correct log-odds, award } \frac{1}{2} \text{ if substituted in correct age but got wrong log-odds)}$$

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x + \dots)}}$$

$$p = \frac{1}{1 + e^{-(-0.45325)}}$$

$$p = 0.39 \text{ (1 for correct probability, award } \frac{1}{2} \text{ if substituted in the log-odds correctly but got wrong solution)}$$

If all done in one step with the correct final answer, award full marks.

- c. You use the resulting model from this analysis to classify the observations in the dataset. A probability threshold of 0.6 is used for the classification. The classification resulted in 43 patients predicted not having a positive response and 57 predicted having a positive response to the medication. Complete the classification matrix shown below by finding the missing values. (2)

		Observed	
		0	1
Predicted	0		5
	1		45

		Observed	
		0	1
Predicted	0	38	5
	1	12	45

1 mark for each correct entry.

If a student does not draw the matrix but still indicates what each value represents you can award full marks. (eg. Tn=38 and fp = 12)

- d. Using the classification matrix above, calculate the sensitivity and specificity of the classification. If you cannot answer question c, choose any two values to use for the missing values in the matrix to answer this question. (2)

$$\text{Sensitivity} = \frac{45}{45+5} = 0.9 \text{ (1)}$$

$$\text{Specificity} = \frac{38}{38+12} = 0.76 \text{ (1)}$$

Mark with error from c.

- e. The predicted probability of a patient reacting positively to the medication is found to be 0.5. Would you classify this patient as reacting positively to the medication or not? Justify your response. (1)

This patient would be classified as not reacting positively to the medication ($\frac{1}{2}$). Probability= 0.5 which is less than the threshold of 0.6 ($\frac{1}{2}$).

- f. Why was logistic regression used for this analysis and not simple linear regression? (1)

The response/dependent variable is binary (1).

Do not accept that the dependent variable is not continuous (must indicate that it is binary).

STA2020F: Important Formulae

SIMPLE LINEAR REGRESSION

(1- α)% confidence interval for $\widehat{\beta}_1$:

$$\widehat{\beta}_1 \pm t_{\frac{\alpha}{2}, n-2} \times s_{\widehat{\beta}_1}$$

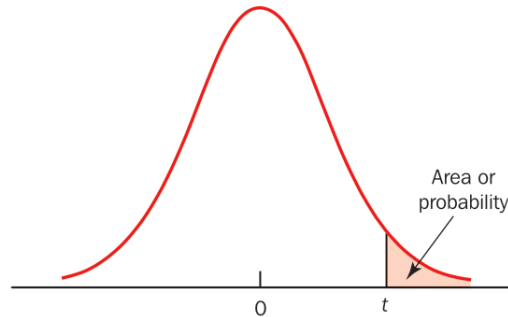
(1- α)% confidence interval for $\widehat{y}|x_p$:

$$\widehat{y} \pm t_{\frac{\alpha}{2}, n-2} \times s_{\epsilon} \times \sqrt{\frac{1}{n} + \frac{(x_p - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

LOGISTIC REGRESSION

Inverse logit:

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x + \dots)}}$$

TABLE 2 *t* distribution

Entries in the table give *t* values for an area or probability in the upper tail of the *t* distribution. For example, with ten degrees of freedom and a .05 area in the upper tail, $t_{.05} = 1.812$.

Degrees of freedom	Area in upper tail					
	.20	.10	.05	.025	.01	.005
1	1.376	3.078	6.314	12.706	31.821	63.656
2	1.061	1.886	2.920	4.303	6.965	9.925
3	.978	1.638	2.353	3.182	4.541	5.841
4	.941	1.533	2.132	2.776	3.747	4.604
5	.920	1.476	2.015	2.571	3.365	4.032
6	.906	1.440	1.943	2.447	3.143	3.707
7	.896	1.415	1.895	2.365	2.998	3.499
8	.889	1.397	1.860	2.306	2.896	3.355
9	.883	1.383	1.833	2.262	2.821	3.250
10	.879	1.372	1.812	2.228	2.764	3.169
11	.876	1.363	1.796	2.201	2.718	3.106
12	.873	1.356	1.782	2.179	2.681	3.055
13	.870	1.350	1.771	2.160	2.650	3.012
14	.868	1.345	1.761	2.145	2.624	2.977
15	.866	1.341	1.753	2.131	2.602	2.947
16	.865	1.337	1.746	2.120	2.583	2.921
17	.863	1.333	1.740	2.110	2.567	2.898
18	.862	1.330	1.734	2.101	2.552	2.878
19	.861	1.328	1.729	2.093	2.539	2.861
20	.860	1.325	1.725	2.086	2.528	2.845
21	.859	1.323	1.721	2.080	2.518	2.831
22	.858	1.321	1.717	2.074	2.508	2.819
23	.858	1.319	1.714	2.069	2.500	2.807
24	.857	1.318	1.711	2.064	2.492	2.797
25	.856	1.316	1.708	2.060	2.485	2.787
26	.856	1.315	1.706	2.056	2.479	2.779
27	.855	1.314	1.703	2.052	2.473	2.771
28	.855	1.313	1.701	2.048	2.467	2.763
29	.854	1.311	1.699	2.045	2.462	2.756

TABLE 2 (Continued)

Degrees of freedom	Area in upper tail					
	.20	.10	.05	.025	.01	.005
30	.854	1.310	1.697	2.042	2.457	2.750
31	.853	1.309	1.696	2.040	2.453	2.744
32	.853	1.309	1.694	2.037	2.449	2.738
33	.853	1.308	1.692	2.035	2.445	2.733
34	.852	1.307	1.691	2.032	2.441	2.728
35	.852	1.306	1.690	2.030	2.438	2.724
36	.852	1.306	1.688	2.028	2.434	2.719
37	.851	1.305	1.687	2.026	2.431	2.715
38	.851	1.304	1.686	2.024	2.429	2.712
39	.851	1.304	1.685	2.023	2.426	2.708
40	.851	1.303	1.684	2.021	2.423	2.704
41	.850	1.303	1.683	2.020	2.421	2.701
42	.850	1.302	1.682	2.018	2.418	2.698
43	.850	1.302	1.681	2.017	2.416	2.695
44	.850	1.301	1.680	2.015	2.414	2.692
45	.850	1.301	1.679	2.014	2.412	2.690
46	.850	1.300	1.679	2.013	2.410	2.687
47	.849	1.300	1.678	2.012	2.408	2.685
48	.849	1.299	1.677	2.011	2.407	2.682
49	.849	1.299	1.677	2.010	2.405	2.680
50	.849	1.299	1.676	2.009	2.403	2.678
51	.849	1.298	1.675	2.008	2.402	2.676
52	.849	1.298	1.675	2.007	2.400	2.674
53	.848	1.298	1.674	2.006	2.399	2.672
54	.848	1.297	1.674	2.005	2.397	2.670
55	.848	1.297	1.673	2.004	2.396	2.668
56	.848	1.297	1.673	2.003	2.395	2.667
57	.848	1.297	1.672	2.002	2.394	2.665
58	.848	1.296	1.672	2.002	2.392	2.663
59	.848	1.296	1.671	2.001	2.391	2.662
60	.848	1.296	1.671	2.000	2.390	2.660
61	.848	1.296	1.670	2.000	2.389	2.659
62	.847	1.295	1.670	1.999	2.388	2.657
63	.847	1.295	1.669	1.998	2.387	2.656
64	.847	1.295	1.669	1.998	2.386	2.655
65	.847	1.295	1.669	1.997	2.385	2.654
66	.847	1.295	1.668	1.997	2.384	2.652
67	.847	1.294	1.668	1.996	2.383	2.651
68	.847	1.294	1.668	1.995	2.382	2.650
69	.847	1.294	1.667	1.995	2.382	2.649

(continued)